

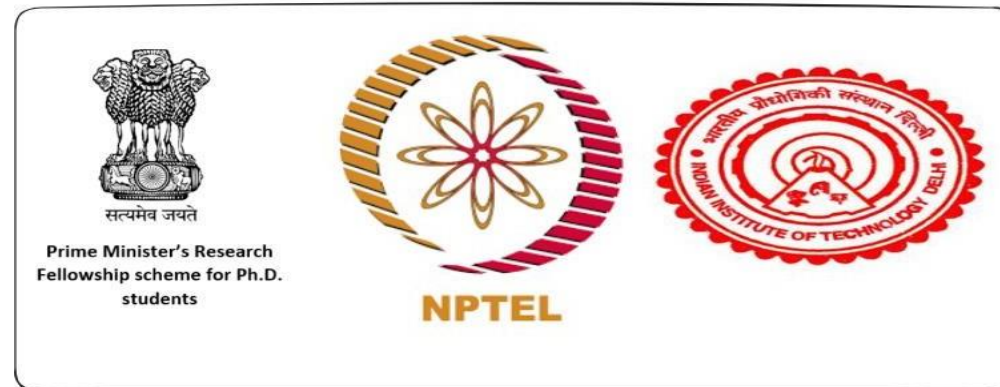
Interaction Session 2

on

Cloud Computing
noc25-cs11

by

Mainak Chakraborty



4th February 2025
Indian Institute of Technology, Delhi

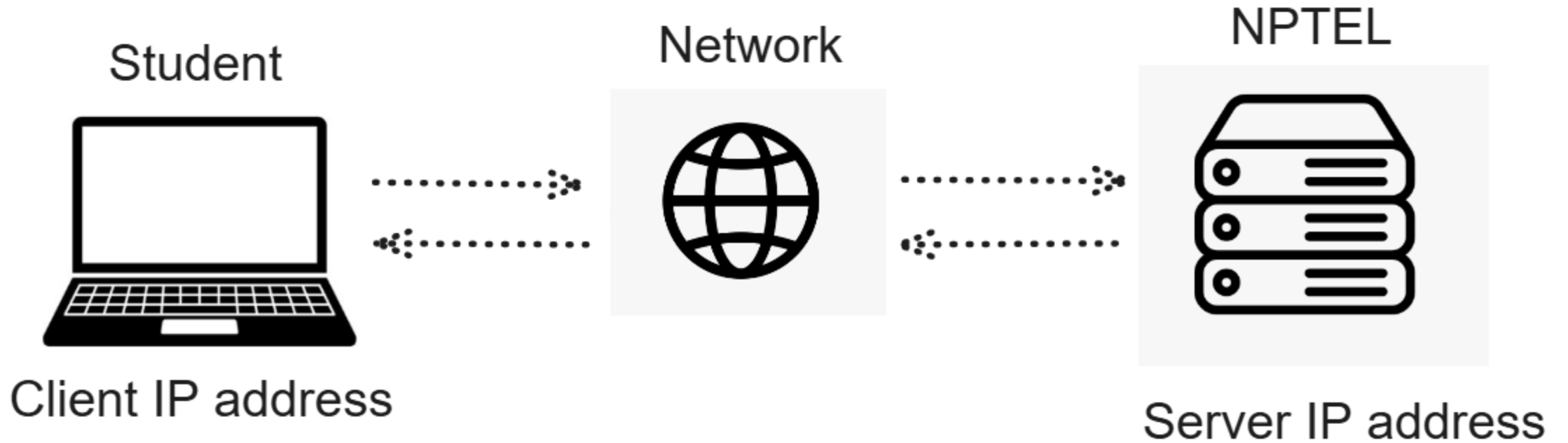
Points to Remember:

- Total 10 more Discussion sessions to be taught lectures and Tutorials
- Any specific problem, please upload in the discussion forum, I will try to reply within 24 hours
- PPT and Video will be shared with everyone
- Assignments will not be solved but similar problems will be solved.

TOPICS TO BE COVERED

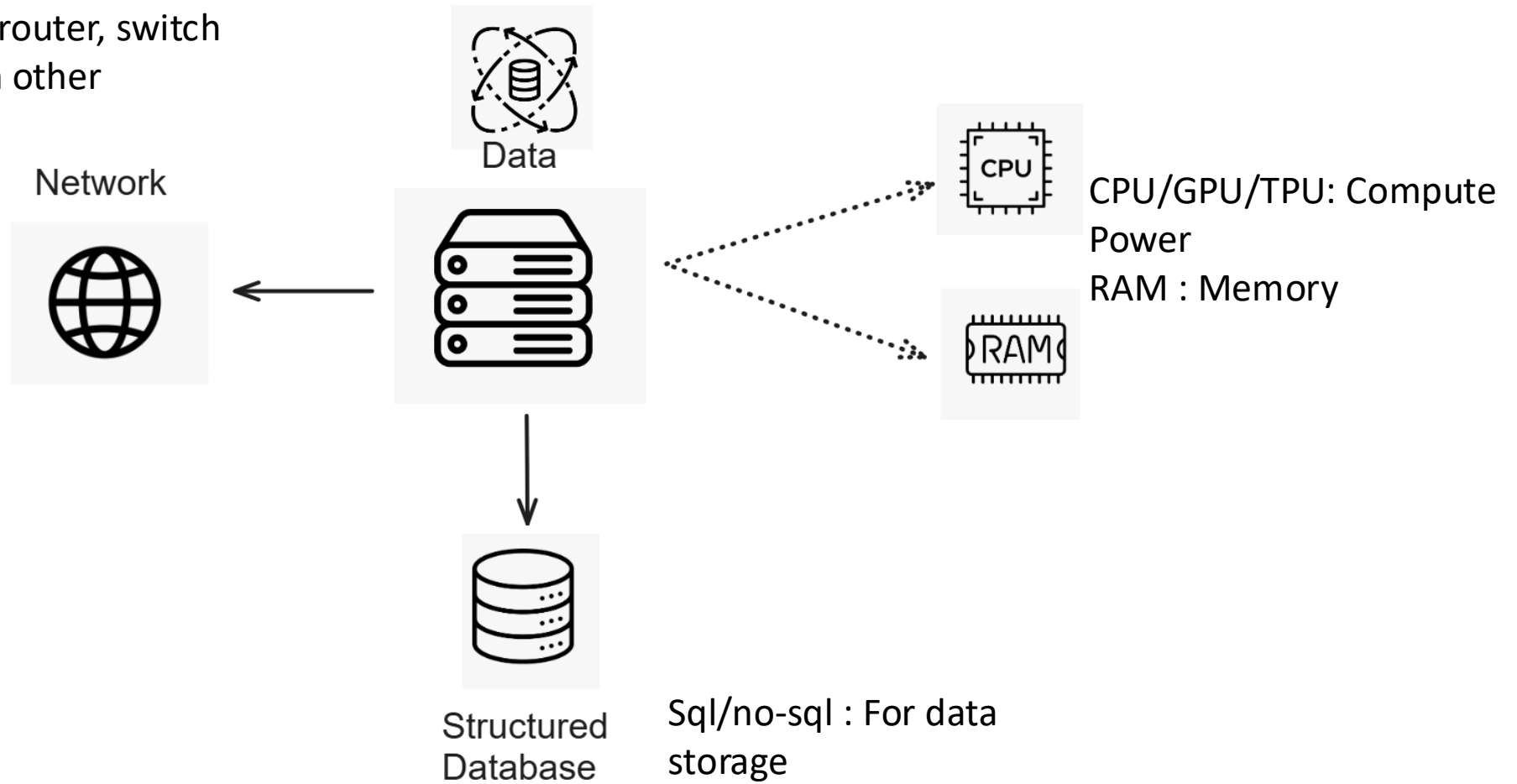
1. Recap
2. Deployment Models
3. Virtualization
4. XML
5. XML-parser using Python
6. Web Services

Background

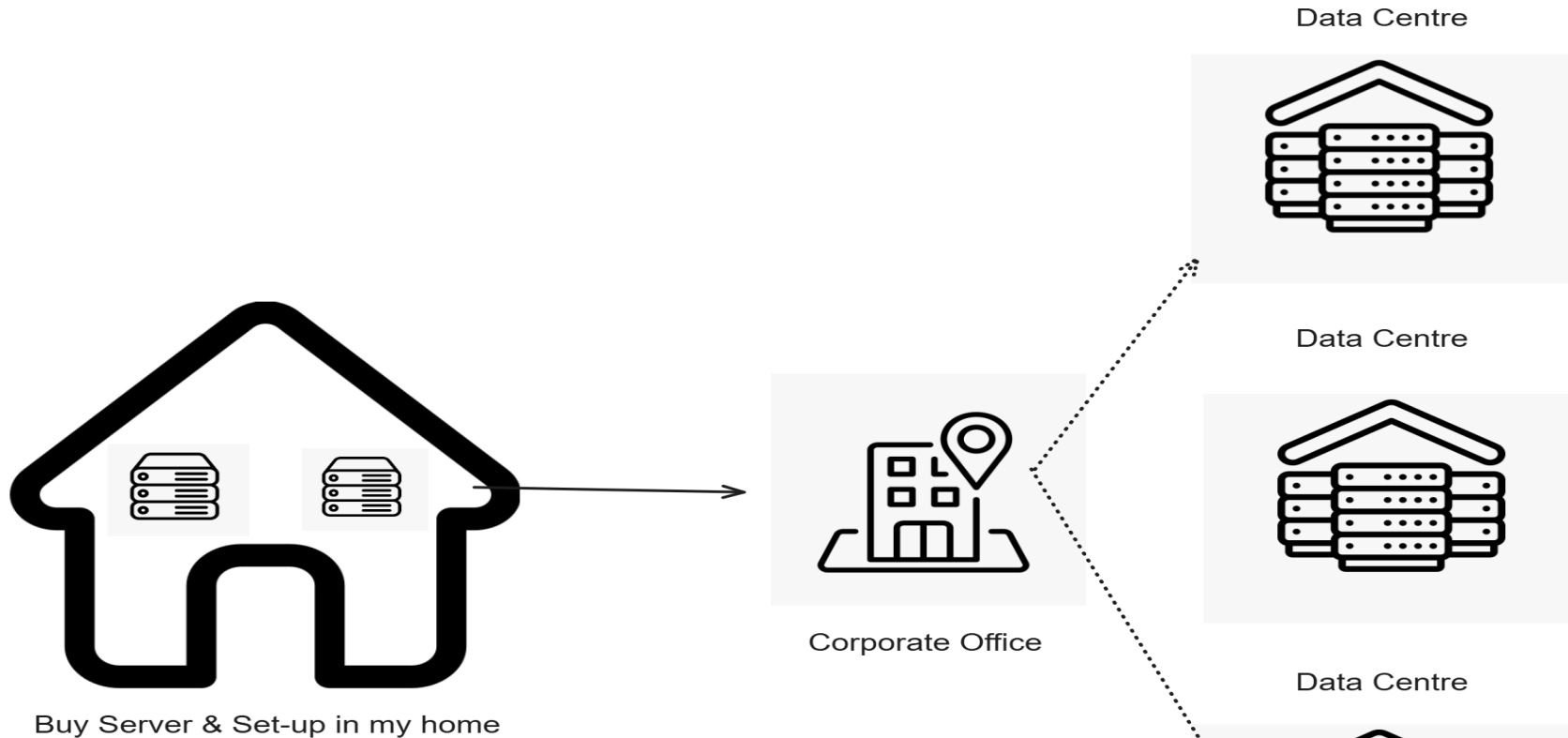


What is the server consisting of?

Network : Cables, router, switch
connected to each other



Traditional IT Approach



Google/Amazon
was made in a garage Which
means they had a server hosted in
their garage...

Example ?

What do we want ?

01

We need "On-Demand" delivery of computation, memory, storage, database

02

We want to pay only for what we use - "Pay-as-you-go" pricing

03

Flexibility / Elasticity to expand and contract. Example: Electricity Supply in our house.

04

Easy-to-use UI for all type of users. Windows/Ubuntu etc.

05

Example : Google Drive, OneDrive, Gmail, Google Photos, Netflix....

Hmmm...but
what if I am a
institution?

Types of Cloud Computing

- **Infrastructure as a Service (IaaS) :** Infrastructure-as-a-Service (IaaS) is a form of cloud computing service that offers compute, storage and networking resources on-demand, usually on a pay-as-you-go basis. Businesses can purchase resources on-demand and as-needed instead of having to buy the hardware outright.

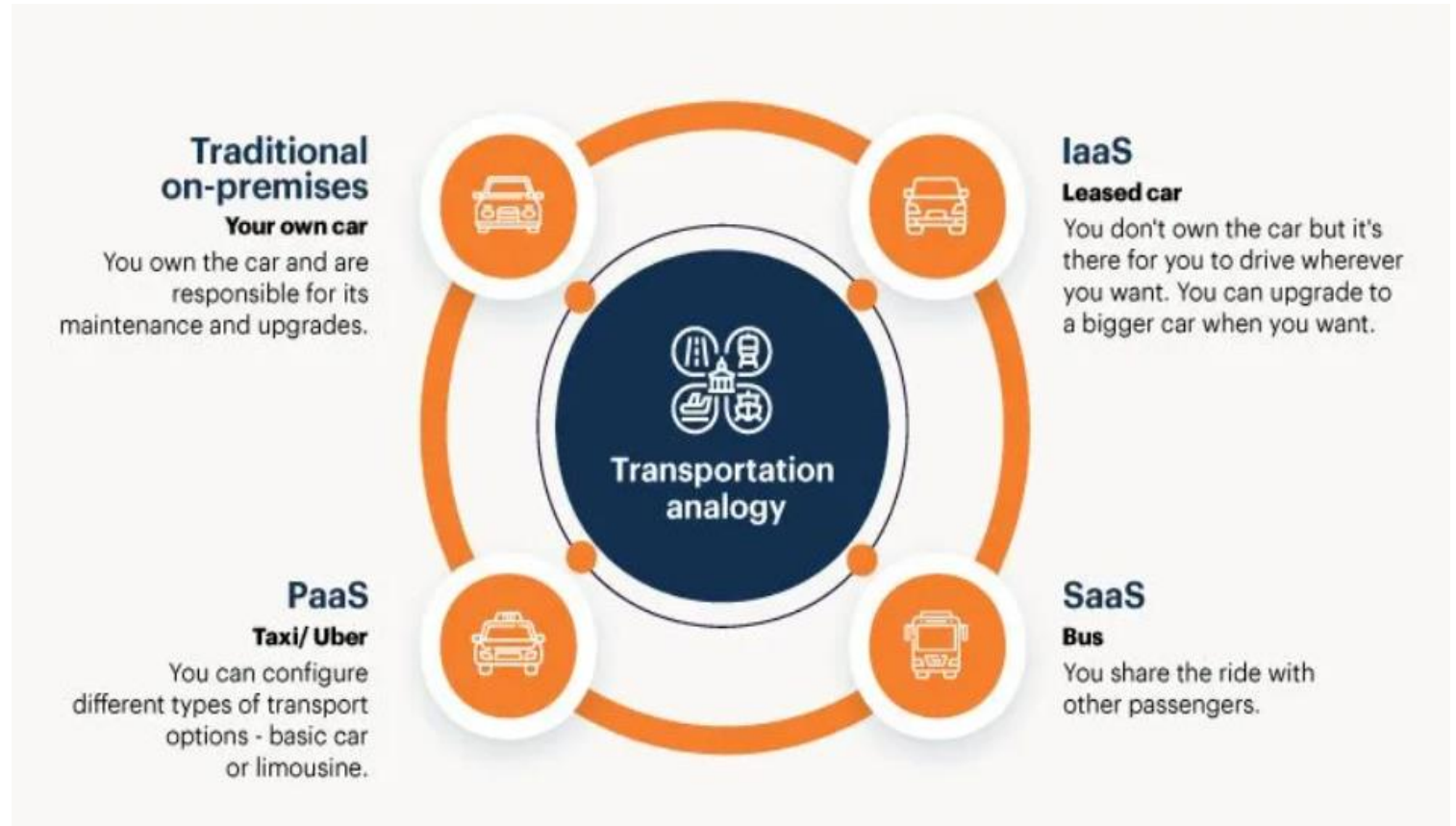
Amazon EC2 (on AWS)

- **Platform as a Service (PaaS) :** Platform-as-a-Service (PaaS) provides a framework for application creation and deployment. The technical stack (e.g., AWS Beanstalk, Google App Engine) required for application development is available on the cloud, which requires no download or local installation.

Windows Azure (Microsoft)

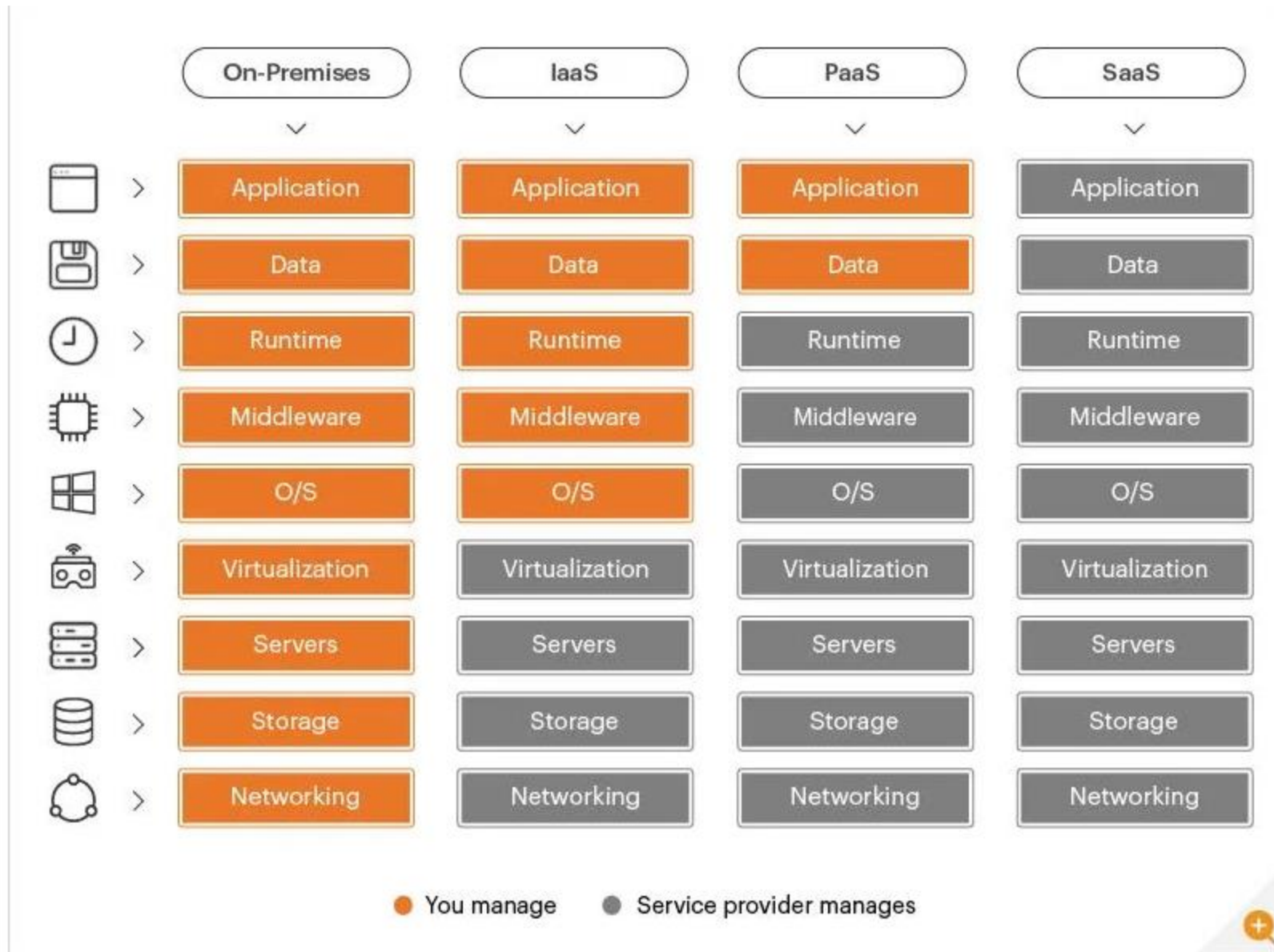
- **Software as a Service (SaaS) :** Software as a Service (SaaS) is a web-based software deployment model that makes the software accessible through a web browser. As a user of SaaS software, you don't care where the software is hosted, which operating system it uses, or which language it is written in.

Google Apps (Gmail)



Source: <https://www.eginnovations.com/blog/saas-vs-paas-vs-iaas-examples-differences-how-to-choose/>

Stack Visualization



The Deployment Models of the Cloud

Private Cloud

- Ubuntu Enterprise Cloud - UEC – Amazon VPC (Virtual Private Cloud)
- IITkgp-->Megh
- Complete Control, specific Business need, not accessible to public

Public Cloud

- Third Party Providers like AWS, Azure, Google Cloud etc.
-

Hybrid Cloud

- Private+Public
- Keep some servers on premises and extend some capabilities to the Cloud

Characteristics

Network, Multi-tenancy

Multi-tenant

Factors	Public Cloud	Private Cloud	Hybrid Cloud
Resources	Resources are shared among multiple customers	Resources are shared with a single organization	It is a combination of public and private clouds. based on the requirement.
Tenancy	Data of multiple organizations is stored in the public cloud	Data of a single organization is stored in a clouds the public cloud	Data is stored in the public cloud, and provide security in the public cloud.
Pay Model	Pay what you used	Have a variety of pricing models	It can include a mix of public cloud pay-as-you-go pricing, and private cloud fixed pricing. It has other pricing models such as consumption-based, subscription-based, etc.
Operated by	Third-party service provider	Specific organization	Can be a combination of both
Scalability and Flexibility	It has more scalability and flexibility,	It has predictability and consistency	It has scalability and flexibility by allowing organizations to use a combination of public and private cloud services.
Expensive	less expensive	More expensive	Can be more expensive, but it can also be less expensive , depending on the specific needs and requirements of the organization.
Availability	The general public (over the internet)	Restricted to a specific organization	Can be a combination of both.

XML

```
<note>
  <to>Student</to>
  <from>NPTEL</from>
  <heading>Reminder</heading>
  <body>Don't Forget WEEKLY Session</body>
</note>
```

XML (Extensible Markup Language) is a markup language similar to [HTML](#), but without predefined tags to use. Instead, you define your own tags designed specifically for your needs. This is a powerful way to store data in a format that can be stored, searched, and shared. Most importantly, since the fundamental format of XML is standardized, if you share or transmit XML across systems or platforms, either locally or over the internet, the recipient can still parse the data due to the standardized XML syntax.

Rules:

1. All XML elements must have a closing tag.
2. XML tags are case-sensitive.
3. All XML elements must be properly nested.
4. All XML documents must have a root element.
5. Attribute values must always be quoted.

```
<?xml version= "1.0" encoding="ISO-8859-1"?>
  <note date="05/05/05">
    <to>Dick</to>
    <from>Jane</from>
  </note>
```

Source: <https://www.ibm.com/docs/hr/b2bis?topic=overview-xml-syntax-rules>

XML Processing with Python

XML-SAX

- It processes the document sequentially and generates events as it encounters tags (start tags, end tags, text).
- Suitable for large XML files as it doesn't load the entire file into memory.
- Low memory footprint

XML-DOM

- DOM-based parser: It reads the entire XML file and constructs an in-memory tree representation of the document.
- Allows for random access to any part of the document tree, making it easy to modify the document.
- High memory usage

XML- element Tree

- Hybrid approach: It builds an element tree but can also be used to iteratively parse large XML files.
- Balances ease of use with performance, suitable for both large and small XML files.
- Easier manipulation and searching through the XML structure compared to `xml.sax`.

Source: <https://www.ibm.com/docs/hr/b2bis?topic=overview-xml-syntax-rules>

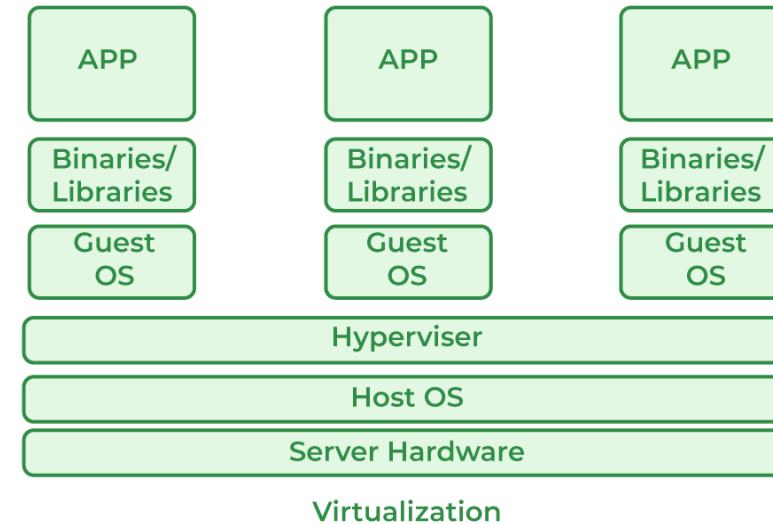
Code : <https://colab.research.google.com/drive/1acDwcm9kZ04wj5UNyT-ARVDSTMrrBge0?usp=sharing>

What is Virtualization?

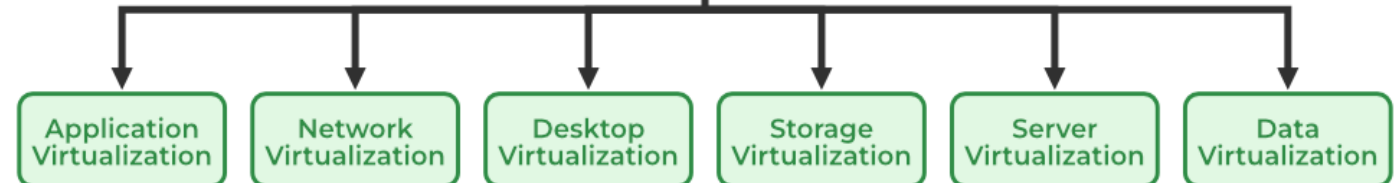
Virtualization is a process that allows a computer to share its hardware resources with multiple digitally separated environments. Each virtualized environment runs within its allocated resources, such as memory, processing power, and storage. With virtualization, organizations can switch between different operating systems on the same server without rebooting.

Virtual machines and hypervisors are two important concepts in virtualization.

dual-boot and virtualization allow you to run multiple operating systems on a single computer, the key difference is that dual-boot lets you only use one operating system at a time by switching between them during boot, whereas virtualization enables you to run multiple operating systems simultaneously within a virtual environment, sharing the host computer's resources.



Types of Virtualization

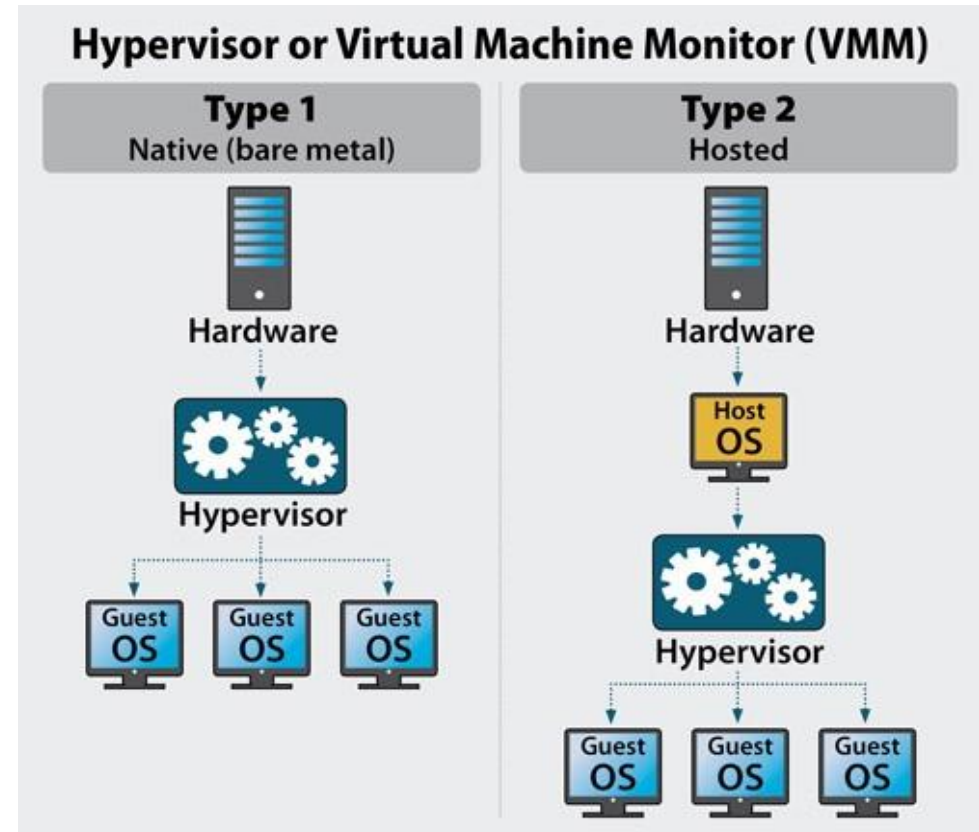


What is a Hypervisor?

The hypervisor (Virtual Machine Monitor) is a software component that manages multiple virtual machines in a computer. It ensures that each virtual machine gets the allocated resources and does not interfere with the operation of other virtual machines. There are two types of hypervisors.

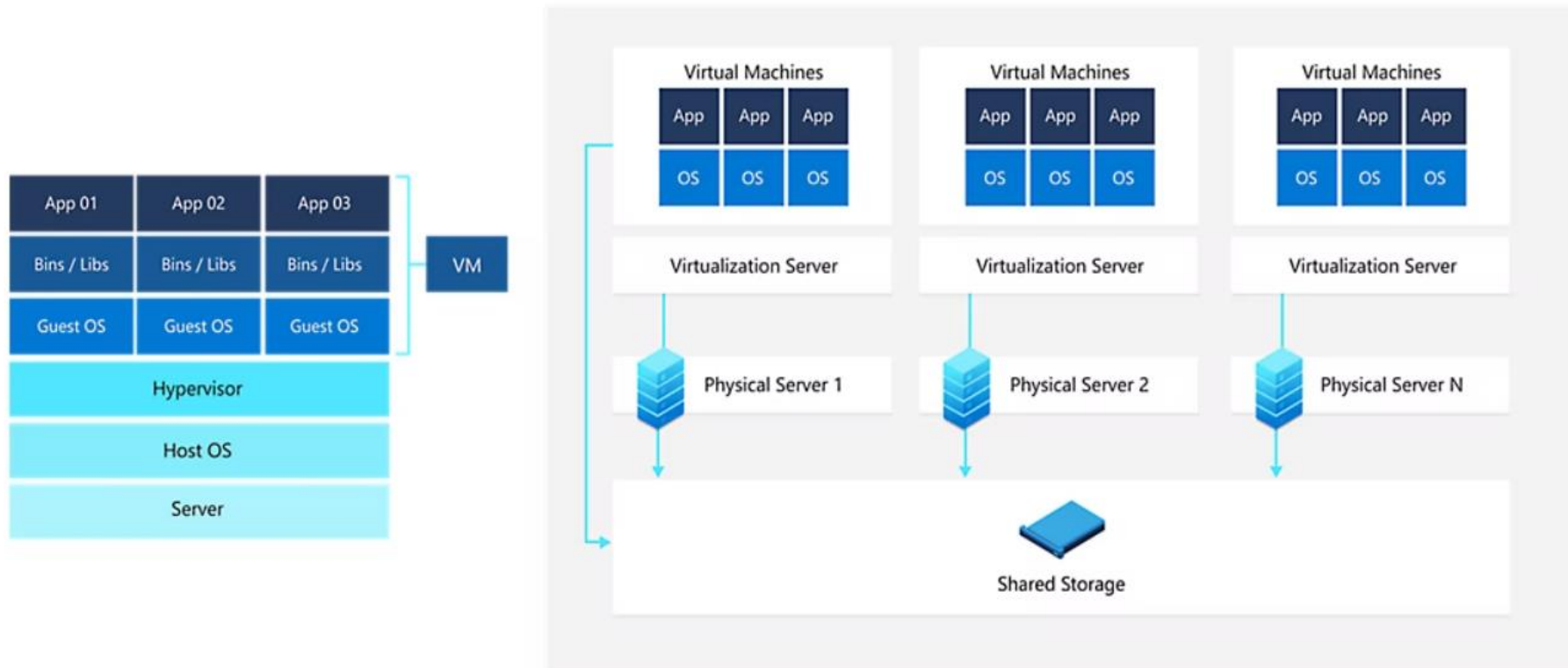
- Type 1 hypervisor : A type 1 hypervisor, or bare-metal hypervisor, is a hypervisor program installed directly on the computer's hardware instead of the operating system. Therefore, type 1 hypervisors have better performance and are commonly used by enterprise applications. KVM uses the type 1 hypervisor to host multiple virtual machines on the Linux operating system.
- Type 2 hypervisor: also known as a hosted hypervisor, the type 2 hypervisor is installed on an operating system. Type 2 hypervisors are suitable for end-user computing.

Source : <https://stl.tech/blog/introduction-to-virtualization/>



What is a Virtual Machine?

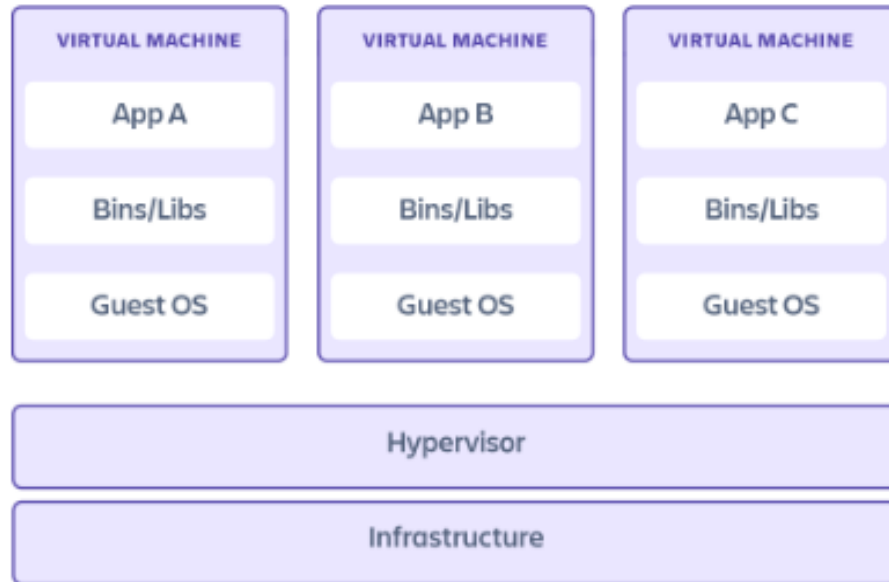
virtual machines run like individual computers with individual operating systems and applications, they have the advantage of remaining completely independent of one another and the physical host machine.



Virtual Machine vs Containerization

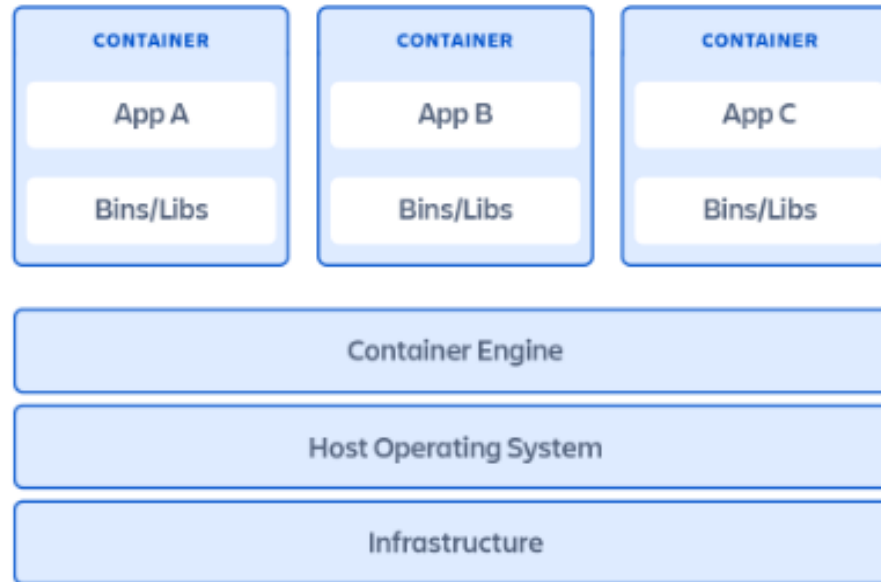
VM is isolation of machines, while Containers is isolation of processes.

Virtual machines



Interacting with machines via hypervisor

Containers

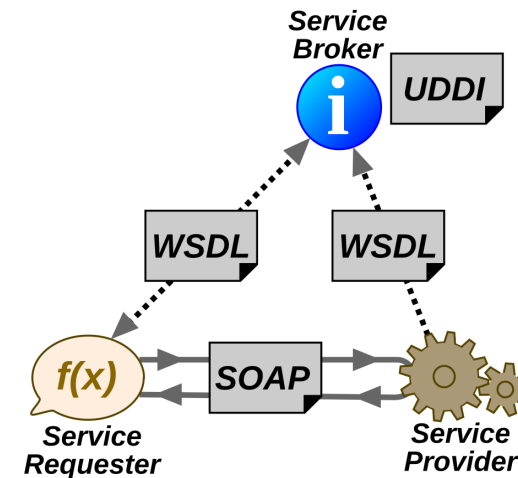


Examples : Docker

What is a "Web service"?

The term "Web service" describes a standardized way of integrating Web-based applications using the XML, SOAP, WSDL and UDDI open standards over an Internet Protocol backbone. XML is the data format used to contain the data and provide metadata around it, SOAP is used to transfer the data, WSDL is used for describing the services available and UDDI lists what services are available.

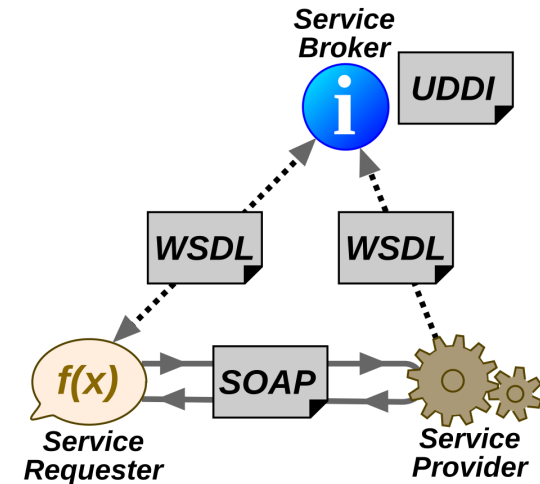
- An application component that:
- Communicates via open protocols (HTTP, SMTP, etc.)
- Processes XML messages framed using SOAP (Simple Object Access Protocol-->Format for sending messages over Internet between programs)
- Describes its messages using XML Schema
- Provides an endpoint description using WSDL
- Can be discovered using UDDI



A Web API is a development in Web services where emphasis has been moving to simpler representational state transfer (REST) based communications.[2] Restful APIs do not require XML-based Web service protocols (SOAP and WSDL) to support their interfaces.

What is a "Web service"?

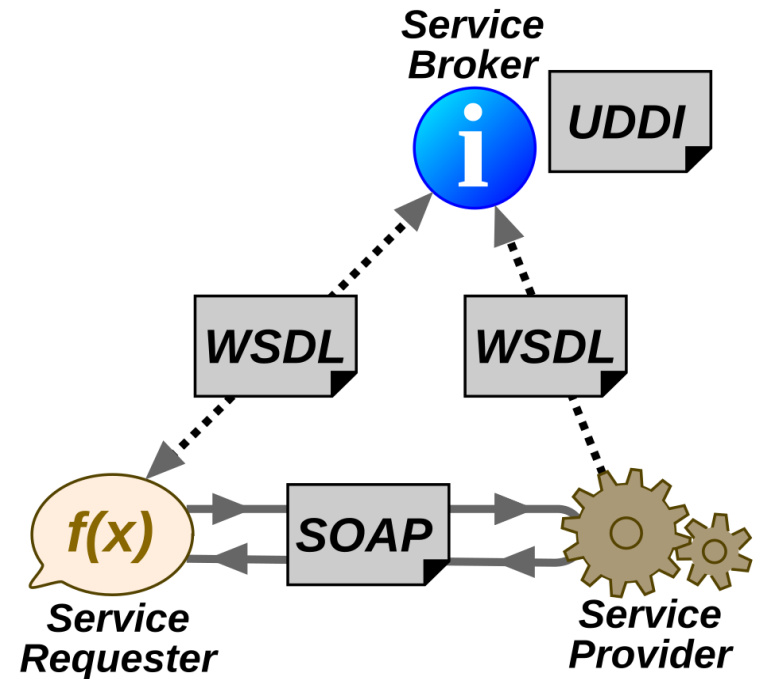
- **Service Requester:** This is the client or consumer that needs a specific service. It interacts with the web service by sending a request.
- **Service Provider:** This is the entity that offers the service. It hosts the web service and makes it available to clients over the network.
- **Service Broker (UDDI):** The Universal Description, Discovery, and Integration (UDDI) acts as a directory service where web services can be registered and categorized. Service providers use UDDI to publish their services, making them discoverable.
- **WSDL (Web Services Description Language):** WSDL is an XML-based language used for describing the functionality offered by a web service. It provides a machine-readable description of how the service can be called, what parameters it expects, and what data structures it returns.
- **SOAP (Simple Object Access Protocol):** SOAP is a protocol used for exchanging structured information in the implementation of web services in computer networks. It uses XML to encode its HTTP-based communications, ensuring that the message structure is understandable by both the service requester and provider.



Interoperability is a key strength of web services. They are designed to enable different systems, applications, and platforms to work together by using standard protocols like HTTP, XML, SOAP, and WSDL. This allows services written in different programming languages and running on different platforms to communicate seamlessly.

What is a "Web service"?

- Discovery:** The service requester queries the UDDI to find a service that meets its requirements. The UDDI responds with a list of services, each described by a WSDL file. The requester selects a service based on the WSDL descriptions.
- Binding:** After selecting a service, the requester uses the information in the WSDL to bind or connect to the service. The WSDL file describes the service endpoints (URLs), the operations that the service exposes, and the parameters and data types these operations accept and return.
- Invocation:** The service requester then constructs a SOAP message, encapsulating the operation to be performed and its required parameters. This SOAP request is sent to the service provider via HTTP.
- Processing:** The service provider receives the SOAP request, processes it according to the operation requested, and returns a SOAP message with the results of the operation.
- Response:** The service requester receives the SOAP response and extracts the data from it for further processing or display.



In essence, while Google search is about finding general information accessible on the web, UDDI is a specialized tool for discovering web services that applications can interact with programmatically for specific business functions.

Hands-on Session

AWS

[Basic Introduction to Services]

AWS Important Points

- Map : https://aws.amazon.com/about-aws/global-infrastructure/regions_az/
- A region is a cluster of data centers
- Most AWS services are region-scoped
- Each availability zone (AZ) is one or more discrete data centers with redundant power, networking, and connectivity
- Pricing Depends on region
- **Account Opening :**
- **AWS Free tier account --> you get 12 months some free services**
https://aws.amazon.com/free/?all-free-tier.sort-by=item.additionalFields.SortRank&all-free-tier.sort-order=asc&awsf.Free%20Tier%20Types=*all&awsf.Free%20Tier%20Categories=*all
- **AWS Educate Account : For course- content you can refer to this**
- BEFORE Opening and using any service- you must know how to close the service and complete information about the services.
- **We will learn about the basics of each services first before we use it next class**

AWS Interface

The screenshot displays the AWS Management Console Home page. At the top, a dark navigation bar includes the AWS logo, a 'Services' menu, a search bar with the placeholder '[Alt+S]', and utility icons for help, notifications, and settings. The user's location is set to 'Mumbai' and their name is 'Deep Patil'. Below the navigation bar, the main content area is titled 'Console Home' and features a 'Reset to default layout' button and an orange '+ Add widgets' button. The left sidebar contains a 'Recently visited' section with a grid of service tiles: Lambda, VPC, AWS Resource Explorer, EC2, EFS, AWS Backup, S3, IAM, CloudShell, DynamoDB, and IAM Identity Center. A 'View all services' link is at the bottom of this list. The right sidebar shows the 'Applications (0)' section for the 'Asia Pacific (Mumbai)' region, with a 'Create application' button and a search bar. Below this is a table header with columns for Name, Description, Region, and Originating account, followed by a message stating 'No applications' and a 'Create application' button. At the bottom of the console, there are three main sections: 'Welcome to AWS', 'AWS Health', and 'Cost and usage'. The footer bar contains links for 'CloudShell', 'Feedback', and copyright information for 2024, along with links for 'Privacy', 'Terms', and 'Cookie preferences'.

aws Services Search [Alt+S] Mumbai Deep Patil

Console Home Info

Reset to default layout + Add widgets

Recently visited Info

- Lambda
- VPC
- AWS Resource Explorer
- EC2
- EFS
- AWS Backup
- S3
- IAM
- CloudShell
- DynamoDB
- IAM Identity Center

View all services

Applications (0) Info

Region: Asia Pacific (Mumbai)

Create application

ap-south-1 (Current Region) Find applications

< 1 >

Name	Description	Region	Originating a...
No applications Get started by creating an application. Create application			

Go to myApplications

Welcome to AWS

AWS Health Info

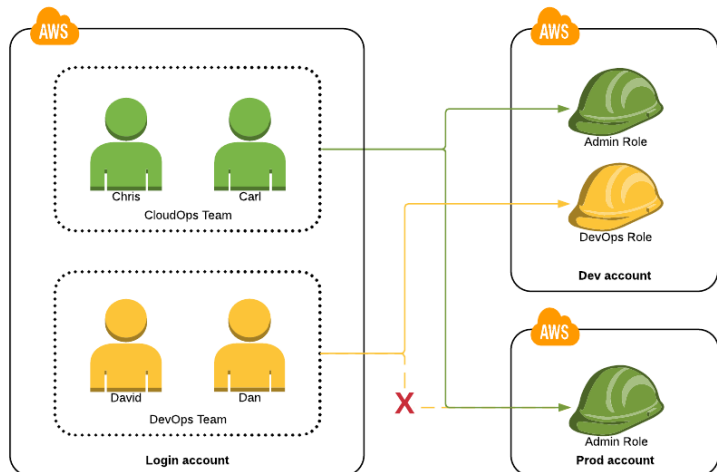
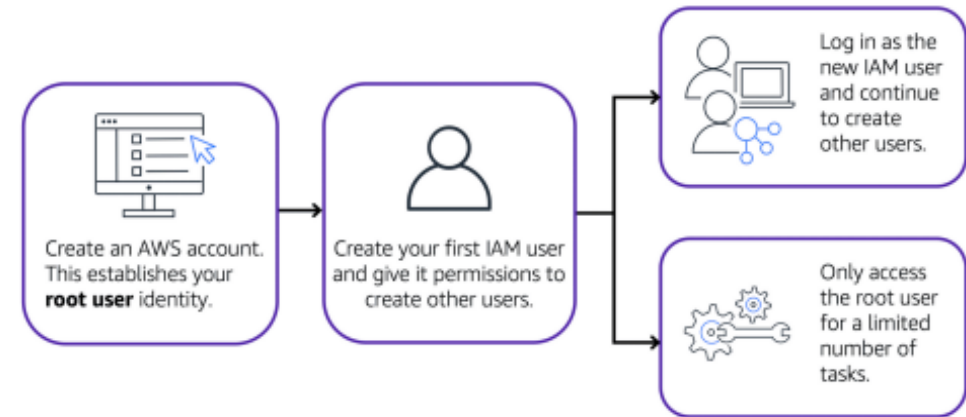
Cost and usage Info

CloudShell Feedback © 2024, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

AWS Account

Your Account's Root User :

- Usage of the root account, especially for daily tasks, introduces several major risks: even a small mistake when using the 'root' user can have catastrophic consequences due to the unrestricted permissions.
- AWS Identity and Access Management **IAM** user



AWS popular Services

Amazon EC2 (Elastic Cloud Compute)

- [Amazon EC2](#) is the fastest cloud computing service provided by AWS. It offers virtual, secure, reliable, and resizable servers for any workload. Through this service, it becomes easy for developers to access resources and also facilitates web-scale cloud computing. This comes with the best suitable processors, networking facilities, and storage systems. Developers can quickly and dynamically scale capacities as per business needs. It has over 500 instances and you can also choose the latest processor, operating system, storage, and networking to help you choose according to the needs of the business. Also, with Amazon EC2, you only have to pay for what you use, and also as per the time period, scale with amazon EC2 auto-scaling has optimal storage and can optimize CPU configurations.

Amazon S3

- With Amazon, it has become easy to store data anytime, anywhere. [Amazon S3 \(Simple Storage Service\)](#), one of the best services provided by AWS is an object storage service offering scalability, availability, security, and high-performing. You can also retrieve data, data here is stored in “storage classes” where there’s no requirement of extra investment and you can also manage it well. **Amazon S3 is the perfect fit for big businesses where a large amount of data is managed for varied purposes.** It comes with handling any volume of data with its robust access controls, and replication tools prevent accidental deletion, and also maintains data version controls.

Amazon Lambda

- Another promising service by AWS is [Amazon Lambda](#) which is a serverless and event-driven computing service that lets you run code for virtual applications or backend services automatically. You need to worry about servers and clusters when working with solutions using Amazon Lambda. It is also cost-effective where you have to only pay for the services you use. **As a user, your responsibility is to just upload the code and Lambda handles the rest.** Using Lambda, you get precise software scaling and extensive availability. With hundreds to thousands of workloads per second, AWS Lambda responsibly handles code execution requests. It is one of the best services provided by AWS for developers.

Amazon EC2 (Elastic Cloud Compute)

An EC2 instance is a virtual server in the AWS Cloud. --> Close

Use Free-tier functions only

Features of Amazon EC2

Amazon EC2 provides the following high-level features:

Instances

Virtual servers.

Amazon Machine Images (AMIs)

Preconfigured templates for your instances that package the components you need for your server (including the operating system and additional software).

Instance types

Various configurations of CPU, memory, storage, networking capacity, and graphics hardware for your instances.

Amazon EBS volumes

Persistent storage volumes for your data using Amazon Elastic Block Store (Amazon EBS).

Instance store volumes

Storage volumes for temporary data that is deleted when you stop, hibernate, or terminate your instance.

Key pairs

Secure login information for your instances. AWS stores the public key and you store the private key in a secure place.

Security groups

A virtual firewall that allows you to specify the protocols, ports, and source IP ranges that can reach your instances, and the destination IP ranges to which your instances can connect

Instance name ▾	On-Demand hourly rate ▲	vCPU ▾	Memory ▾	Storage ▾	Network performance ▾
t4g.nano	\$0.0042	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3a.nano	\$0.0047	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t3.nano	\$0.0052	2	0.5 GiB	EBS Only	Up to 5 Gigabit
t2.nano	\$0.0058	1	0.5 GiB	EBS Only	Low
t4g.micro	\$0.0084	2	1 GiB	EBS Only	Up to 5 Gigabit
t3a.micro	\$0.0094	2	1 GiB	EBS Only	Up to 5 Gigabit
t3.micro	\$0.0104	2	1 GiB	EBS Only	Up to 5 Gigabit
t2.micro	\$0.0116	1	1 GiB	EBS Only	Low to Moderate
t4g.small	\$0.0168	2	2 GiB	EBS Only	Up to 5 Gigabit

<https://aws.amazon.com/ec2/pricing/on-demand/>

<https://www.geeksforgeeks.org/amazon-ec2-creating-an-elastic-cloud-compute-instance/>

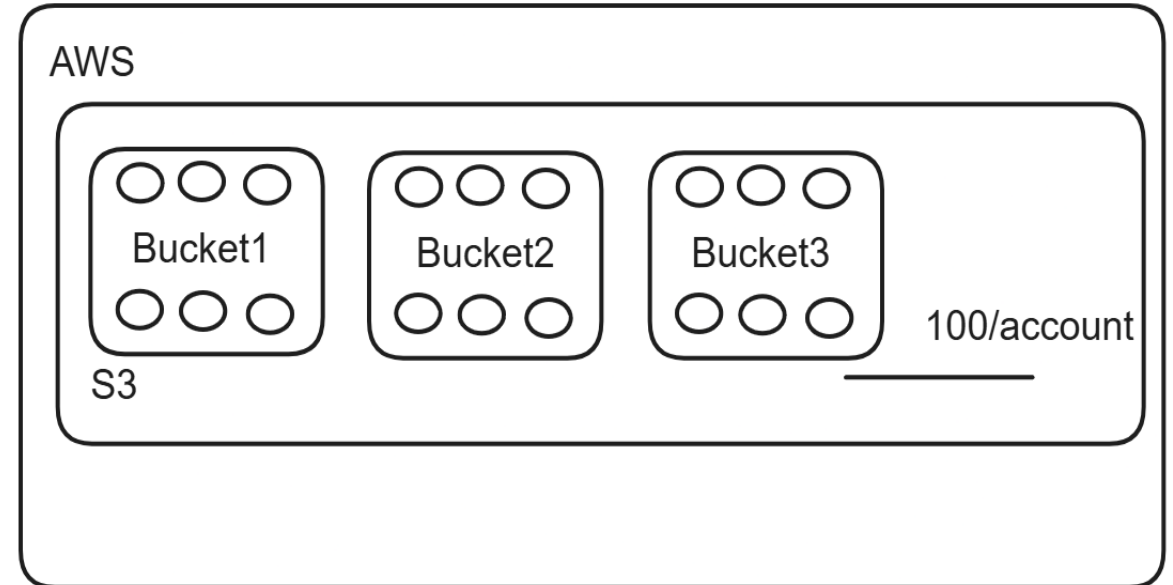
Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.

Use Free-tier
functions only

Features of Amazon S3

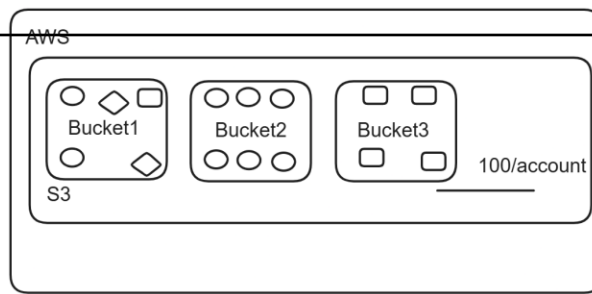
- **Bucket** : A general-purpose bucket is a container for objects stored in Amazon S3. You can store any number of objects in a bucket and all accounts have a default bucket quota of 10,000 general purpose buckets. Every object is contained in a bucket. For example, if the object named `photos/puppy.jpg` is stored in the `amzn-s3-demo-bucket` bucket in the US West (Oregon) Region, then it is addressable by using the URL <https://amzn-s3-demo-bucket.s3.us-west-2.amazonaws.com/photos/puppy.jpg>.
- When you create a bucket, you enter a bucket name and choose the AWS Region where the bucket will reside. After you create a bucket, you cannot change the name of the bucket or its Region. Bucket names must follow the [bucket naming rules](#). You can also configure a bucket to use [S3 Versioning](#) or other [storage management](#) features.
- **Object** : Objects are the fundamental entities stored in Amazon S3. Objects consist of object data and metadata. The metadata is a set of name-value pairs that describe the object. These pairs include some default metadata, such as the date last modified, and standard HTTP metadata, such as Content-Type. You can also specify custom metadata at the time that the object is stored.



<https://aws.amazon.com/ec2/pricing/on-demand/>

Buckets

Amazon Simple Storage Service (Amazon S3) is an object storage service offering industry-leading scalability, data availability, security, and performance.



Use Free-tier
functions only

Features of Buckets

- When you create a general purpose bucket, you choose its name and the AWS Region to create it in. After you create a general purpose bucket, you can't change its name or Region. The following sections provide information about general purpose bucket naming, including naming rules, best practices, and an example for creating a general purpose bucket with a name that includes a globally unique identifier (GUID).
- Important Rules :
 - Bucket names **must** be unique across all AWS accounts in all the AWS Regions within a partition. A partition is a grouping of Regions. AWS currently has three partitions: aws (commercial Regions), aws-cn (China Regions), and aws-us-gov (AWS GovCloud (US) Regions).
 - A bucket name can't be used by another AWS account in the same partition until the bucket is deleted. **After you delete a bucket, be aware that another AWS account in the same partition can use the same bucket name for a new bucket and can therefore potentially receive requests intended for the deleted bucket.** If you want to prevent this, or if you want to continue to use the same bucket name, don't delete the bucket. We recommend that you empty the bucket and keep it, and instead, block any bucket requests as needed.

• Bucket policy

• First delete objects and then delete Bucket

• <https://aws.amazon.com/s3/pricing/>

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Principal": {
        "AWS": "arn:aws:iam::111122223333:role/JohnDoe"
      },
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:GetObjectVersion"
      ],
      "Resource": "arn:aws:s3:::amzn-s3-demo-bucket/*",
      "Condition": {
        "StringEquals": {
          "s3:ExistingObjectTag/environment": "production"
        }
      }
    }
  ]
}
```

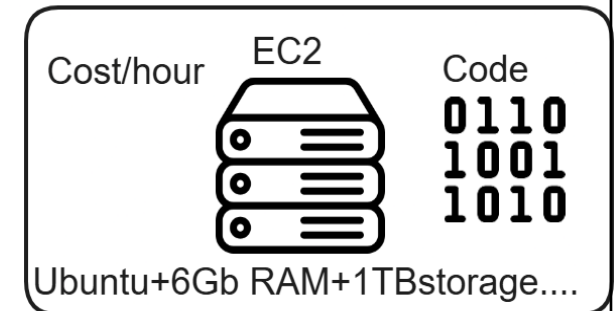
<https://docs.aws.amazon.com/AmazonS3/latest/userguide/Welcome.html#BasicsBucket>

Amazon Lambda: Function as a service (FaaS)

Use Free-tier functions only

AWS Lambda is a serverless computing service that automatically runs code in response to events, without requiring you to manage the underlying infrastructure. It supports event-driven applications triggered by events such as HTTP requests, DynamoDB table updates, or state transitions. You simply upload your code (as a .zip file or container image), and Lambda handles everything from provisioning to scaling and maintenance. It automatically scales applications based on traffic, handling server management, auto-scaling, security patching, and monitoring. AWS Lambda is ideal for developers who want to focus on writing code without worrying about infrastructure management.

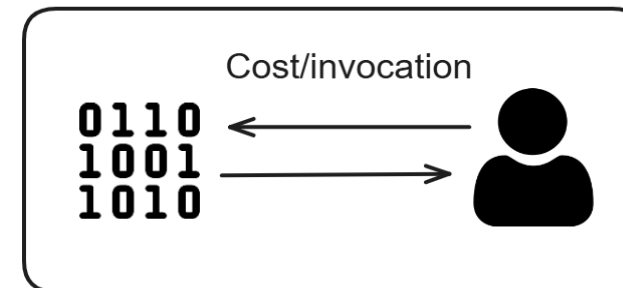
SERVER-->



Features of Amazon Lambda

- AutoScaling and High Availability:
- Serverless Execution:
- Pay-per-use-pricing:
- Supports different programming languages:
- Integrates with other AWS Services:
- Versioning and Deployment:
- Security and Identity Management:

SERVER-Less-->



<https://aws.amazon.com/lambda/pricing/>

Case Study : Food Delivery Startup

- Let's assume you are a mobile app developer building a food ordering app. Customers can use the app to order food from a specific restaurant location, receive order status updates, and pick up the food when the order is ready. Because you expect high variation in demand, both by time of day and restaurant location, you build your mobile backend using serverless services, including AWS Lambda.
- For simplicity, let's assume your application processes three million requests per month. The average function execution duration is 120 ms. You have configured your function with 1536 MB of memory, on an x86 based processor. Your charges would be calculated as follows:
- Monthly compute charges
- The monthly compute price is \$0.0000166667 per GB-s and the free tier provides 400,000 GB-s.
Total compute (seconds) = 3 million * 120ms = 360,000 seconds
Total compute (GB-s) = 360,000 * 1536MB/1024 MB = 540,000 GB-s
Total compute – Free tier compute = monthly billable compute GB- s
540,000 GB-s – 400,000 free tier GB-s = 140,000 GB-s
Monthly compute charges = 140,000 * \$0.0000166667 = \$2.33
- Monthly request charges
- The monthly request price is \$0.20 per one million requests and the free tier provides 1 million requests per month.
Total requests – Free tier requests = monthly billable requests
3 million requests – 1 million free tier requests = 2 million monthly billable requests
Monthly request charges = 2M * \$0.2/M = \$0.40
- Total monthly charges
- Total charges = Compute charges + Request charges = \$2.33 + \$0.40 = \$2.73 per month